

What is Solid Waste?

Solid waste comprises of all the wastes arising from human and animal activities that are typically solid and that are discarded as **useless** or **unwanted**. It is all-inclusive of the **heterogeneous mass** from the urban community as well as more **homogeneous accumulation** of agriculture and industrial wastes.

Some of the commonly used synonyms for solid wastes are:

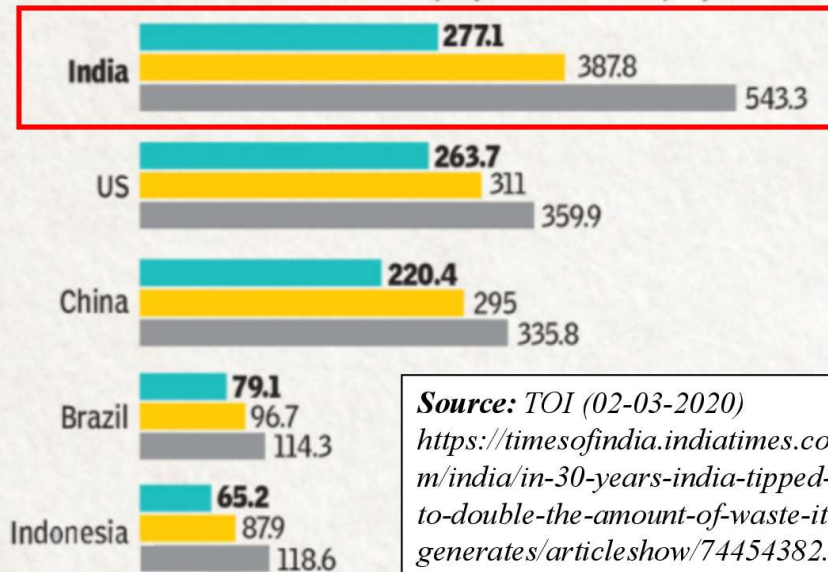
- Refuse: This is a more appropriate term for solid waste as most waste can be utilized as a raw material for some other purpose.
- Garbage: It consist of kitchen/wet waste.
- Rubbish: Those wastes with high ash content.
- Scrap: Wastes that have high metal content.
- Debris: Bulky wastes such as construction waste.

MUNICIPAL SOLID WASTE (MSW) GENERATION

TOP 5 MUNICIPAL SOLID WASTE GENERATORS ANNUALLY

(in million tonnes)

■ 2016 estimate ■ 2030 projected ■ 2050 projected

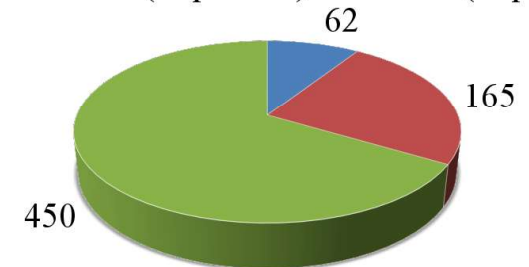


Source: TOI (02-03-2020)
<https://timesofindia.indiatimes.com/india/in-30-years-india-tipped-to-double-the-amount-of-waste-it-generates/articleshow/74454382.cms>

More than tenth of the world's waste is generated in India

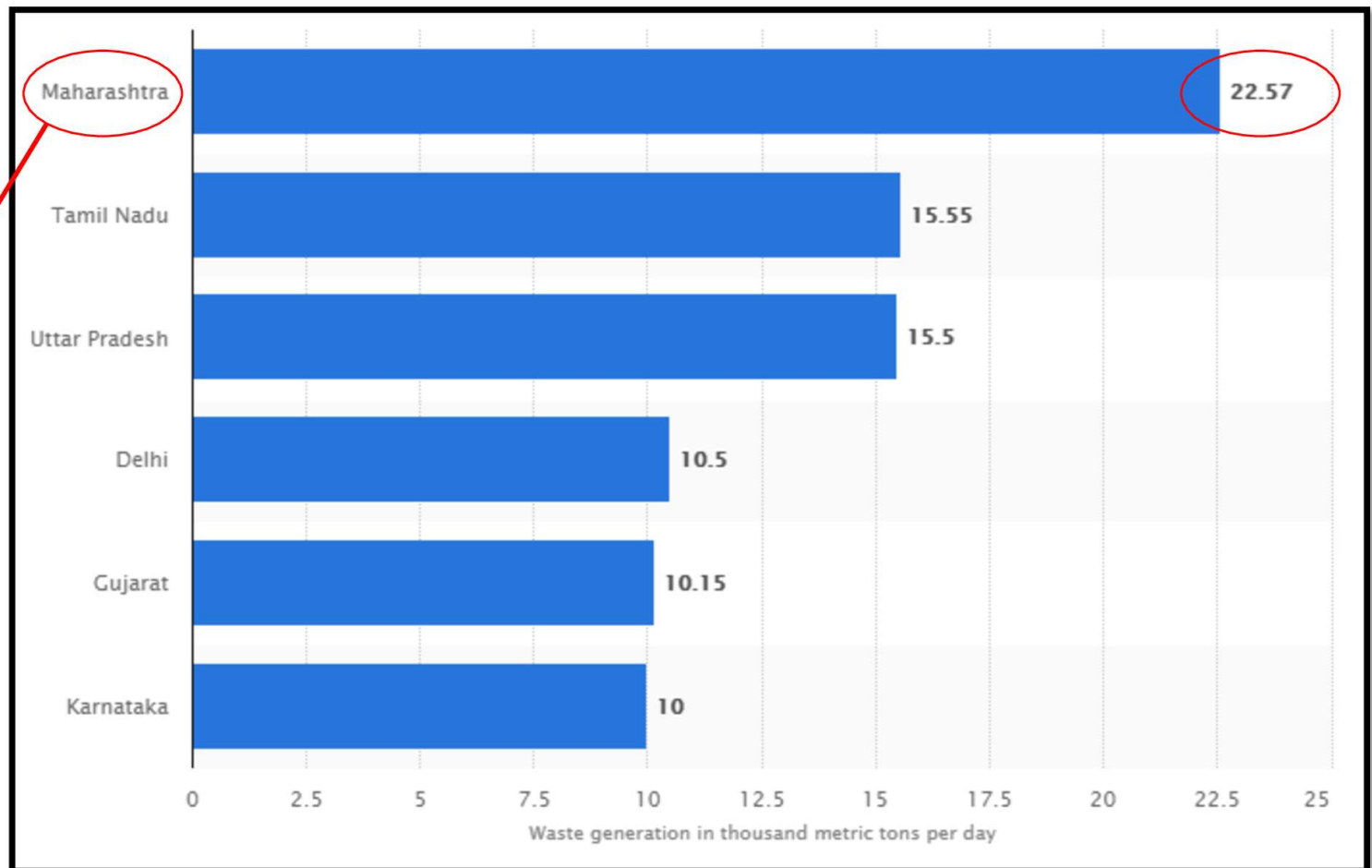
Waste generation in India (million tons)

■ 2017 ■ 2030 (expected) ■ 2045 (expected)



Source: The Hindu (28-04-18)

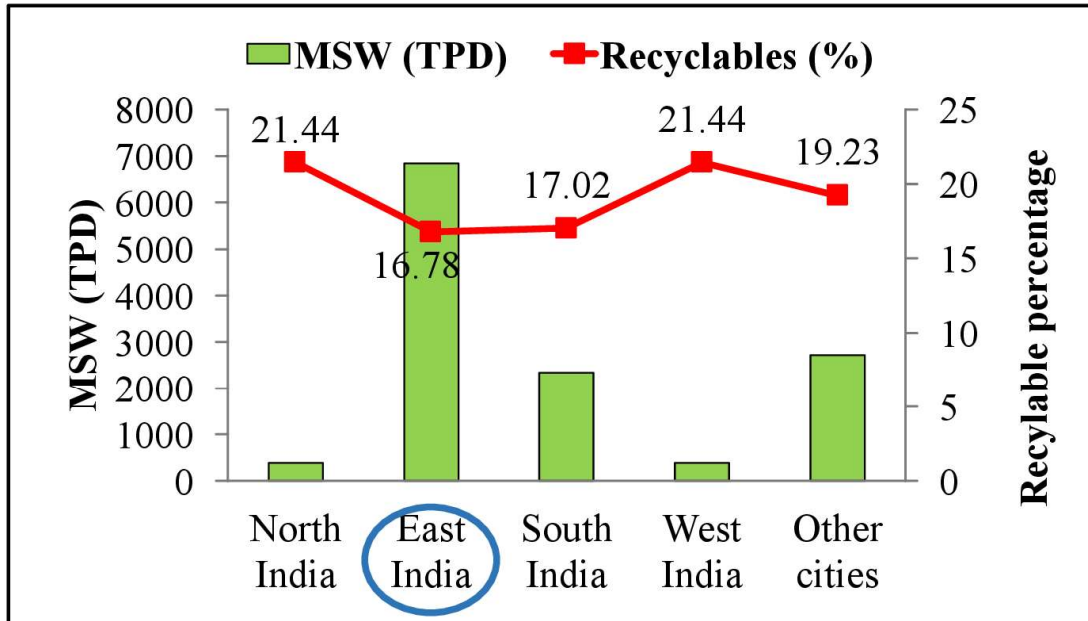
MAJOR SOLID WASTE GENERATING STATES ACROSS INDIA



Generates highest quantity of solid waste among all the states of India as **on November, 2017**

Source: Statistica 2020

Region-wise MSW generation in India

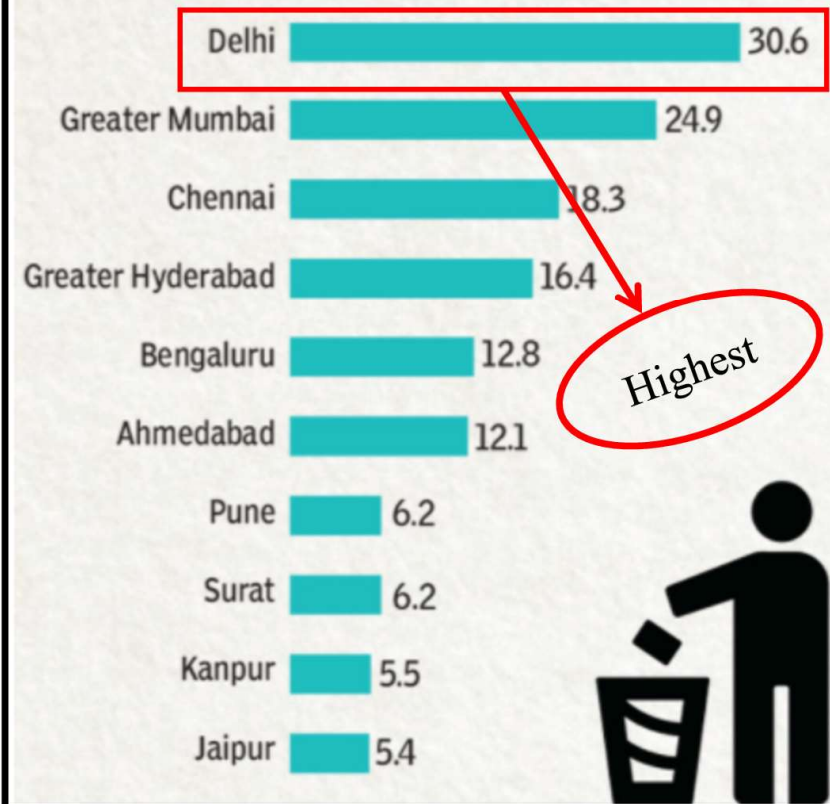


Source: Annepu, R K (2012): Sustainable solid waste management in India, Waste-to-Energy Research and Technology Council (WTER), Columbia University, Earth Engineering Center.

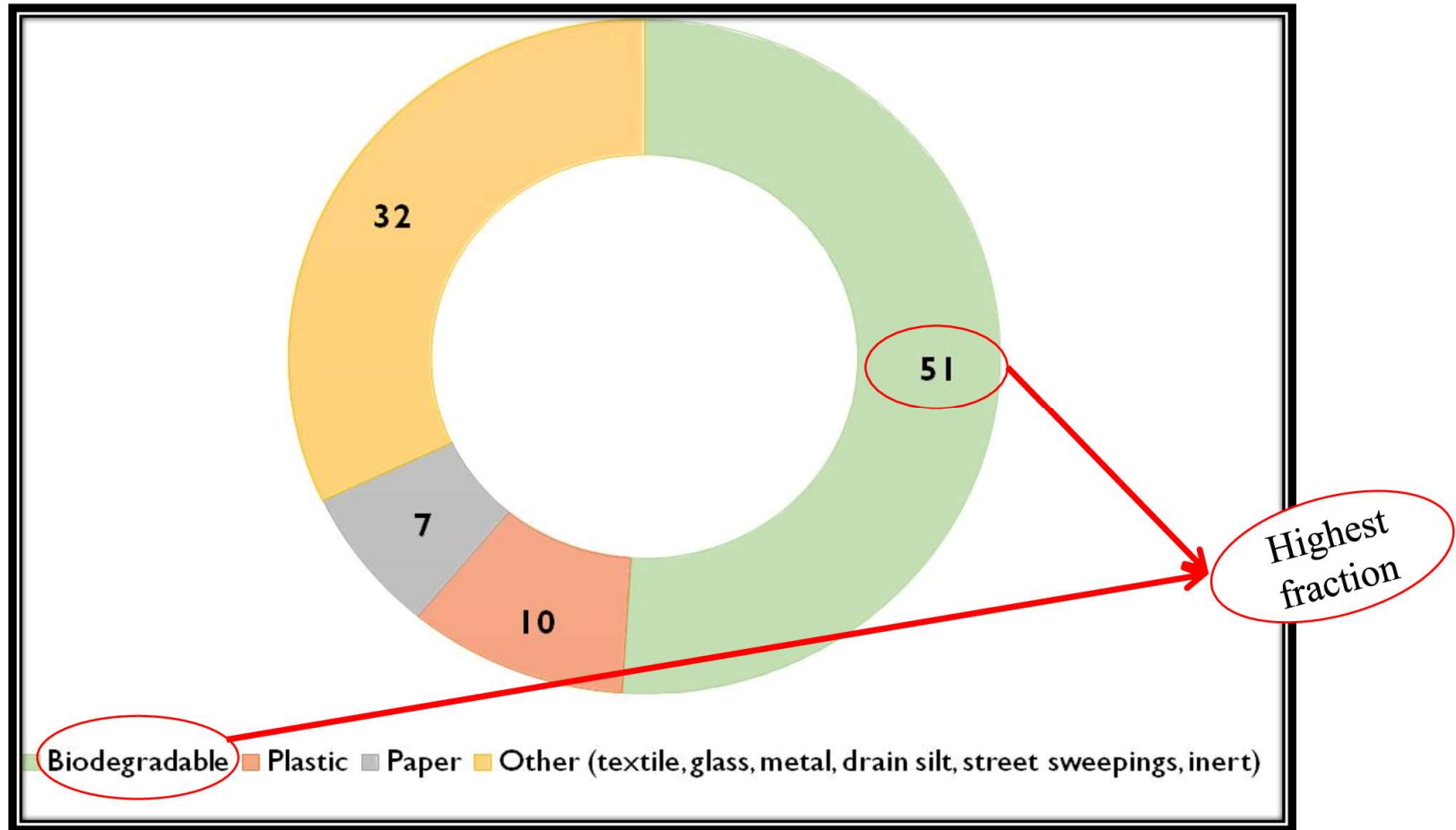
Source: TOI (02-03-2020)
<https://timesofindia.indiatimes.com/india/in-30-years-india-tipped-to-double-the-amount-of-waste-it-generates/articleshow/74454382.cms>

TOP 10 INDIAN CITIES FOR WASTE GENERATION ANNUALLY

(in lakh tonnes)

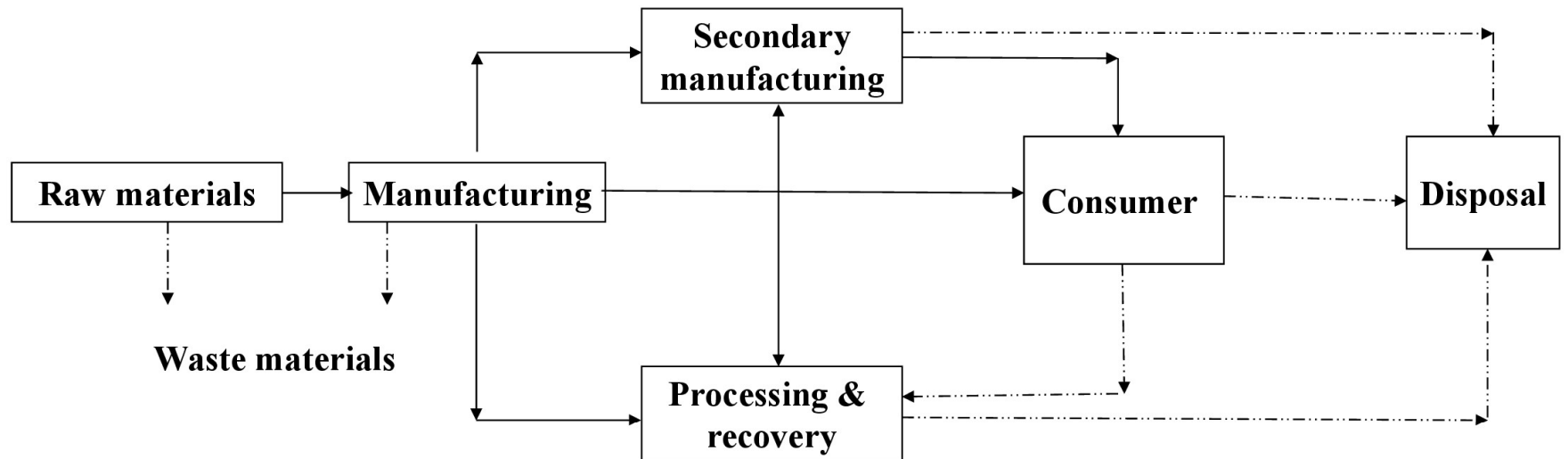


MSW COMPOSITION IN INDIA



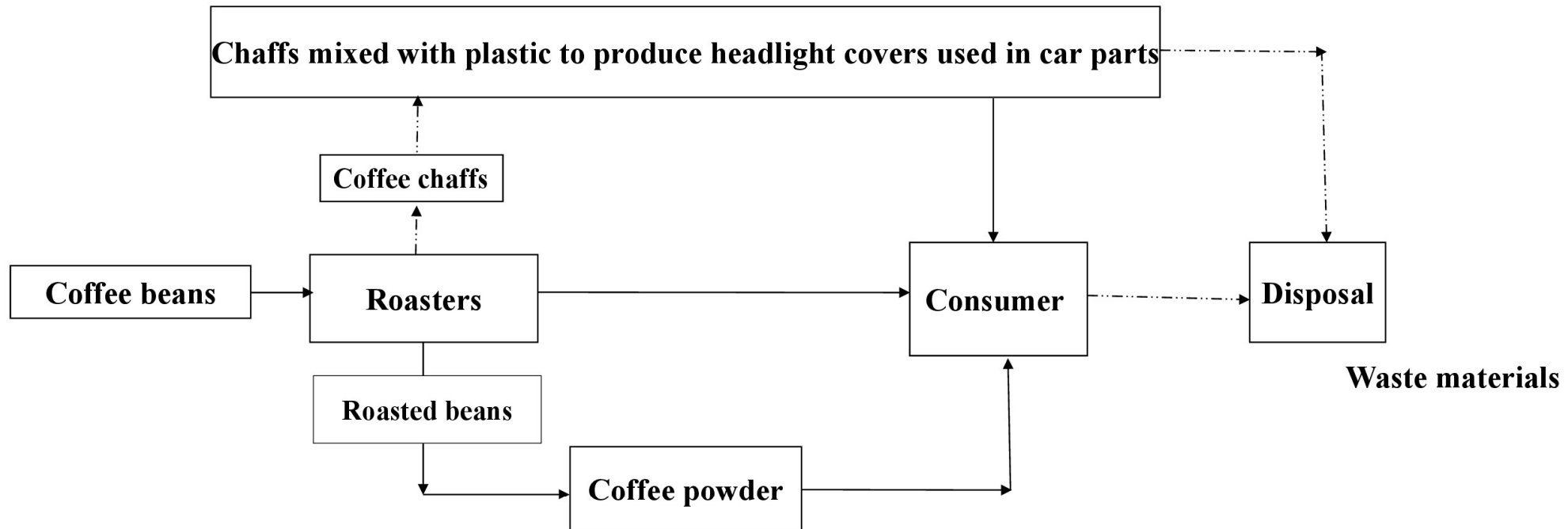
Source: Task Force on Waste to Energy, Planning Commission, 2014

Materials Flow And Waste Generation



-----> waste

Materials Flow And Waste Generation (contd.)



-----> waste

Why solid waste: A consequence of life

- The relationship between public health and improper storage, collection and disposal of solid waste is quite clear.
- The consequences of improper disposal of solid waste can be very well highlighted with the following examples.



Improper disposal causes **plague** because of breeding of rats, flies and other disease vectors.



Improper landfilling of solid waste attributes to **water pollution**.



Improper landfilling of solid waste attributes to air pollution by release of **greenhouse gases**. In Fig: Kolkata's permanently burning landfill.



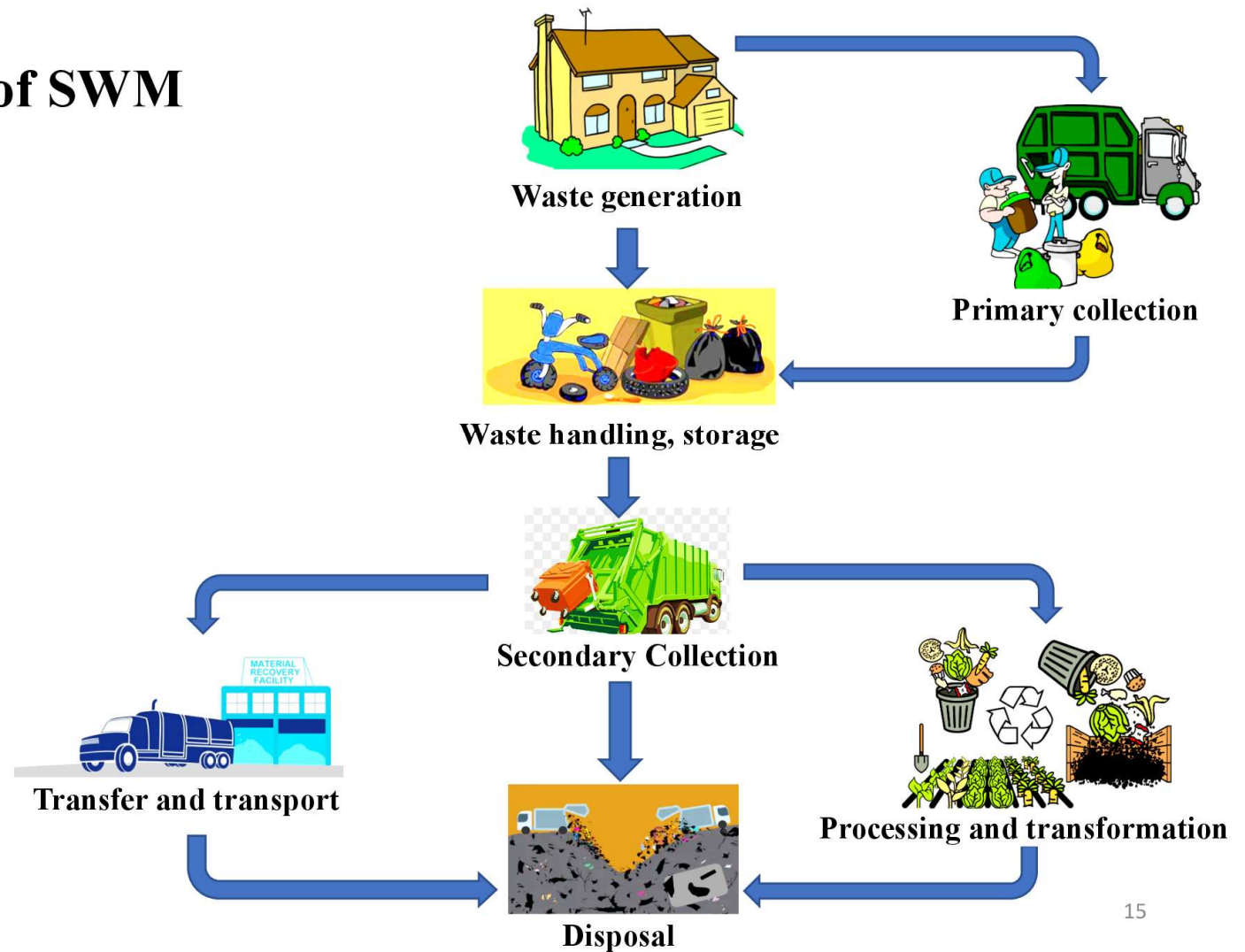
Plastic dumped in ocean causes harm to marine species, threatens food safety and quality, human health, coastal tourism, and contributes to climate change.



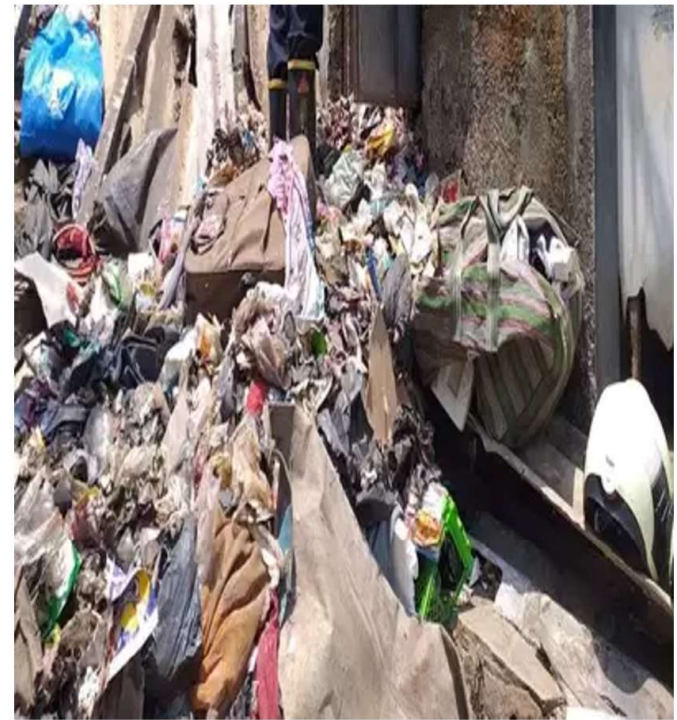
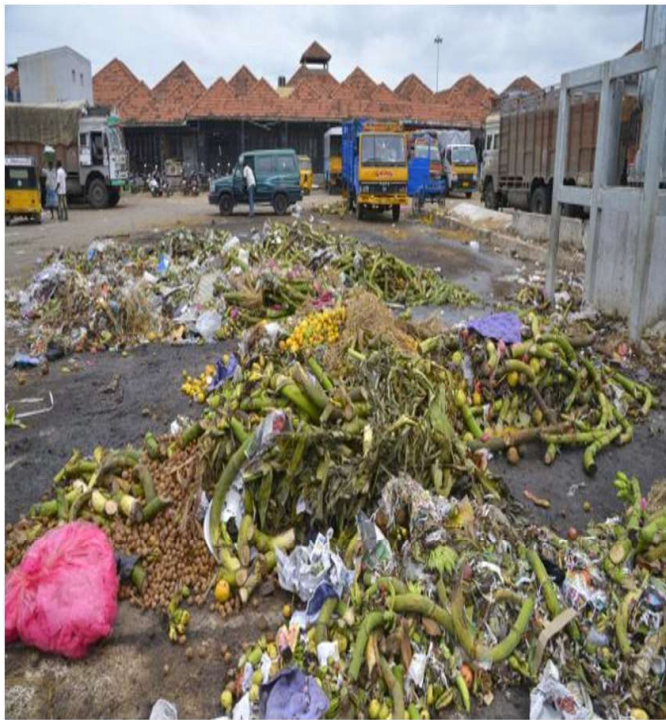
The effects of technological advances: Increasing use of plastics, packaging of goods, use of frozen foods.

Functional Elements of SWM

The activities associated with the management of solid waste from the point of **generation** to **final disposal** have been grouped into the **six functional elements**.



1. **Waste generation:** Waste generation encompasses activities in which materials are identified as no longer being of value and are either thrown away or gathered together for disposal.



2. Waste handling and storage: They involve the activities associated with management of wastes until they are placed in storage container for collection.



HAUL CONTAINER



STATIONARY CONTAINER

3.1 Primary Collection: It includes the gathering of solid wastes and recyclable materials from the sources. but also the operation of collection vehicle to landfill disposal sites.



COMMUNITY LEVEL COLLECTION

OR



DOOR TO DOOR COLLECTION

3.2 Secondary Collection: It includes operation of collection vehicle from dustbin area to landfill disposal sites.



WASTE HAULING



CONTAINER HAULING

4. Processing and transformation of solid wastes: It is a very important function of solid waste management but is often ignored .



RECYCLING OR TRANSFORMATION

5. Transfer and transport: Two steps:

- a) The transfer of wastes from the smaller collection vehicle to the larger transport equipment
- b) The subsequent transport of the wastes to disposal sites.



TRANSFER STATION

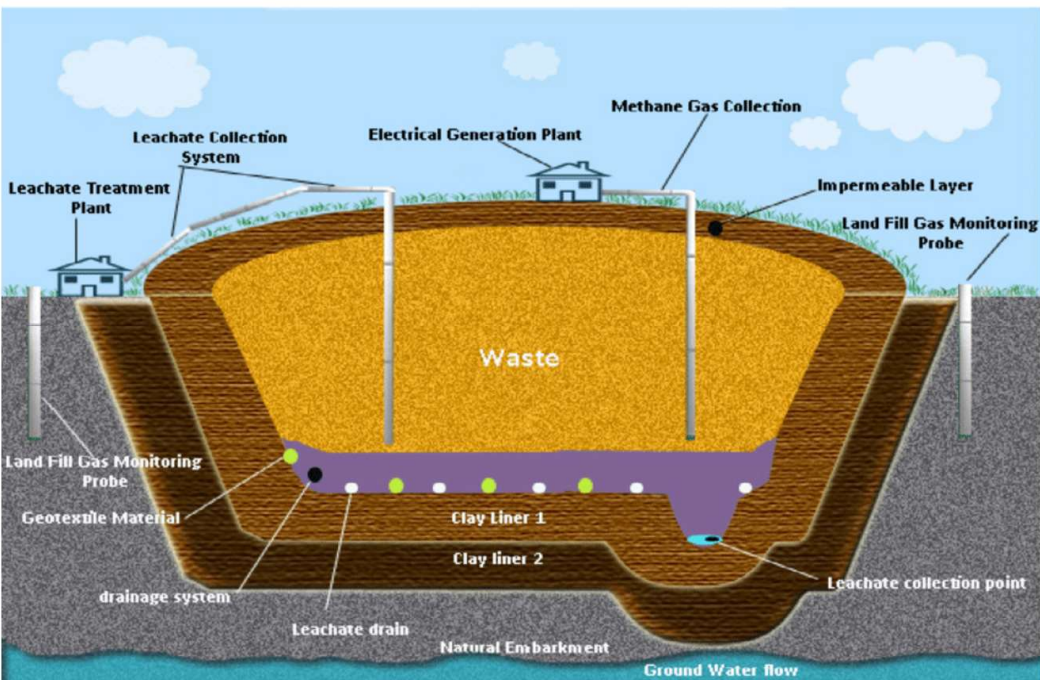


PNEUMATIC REFUSE COLLECTION
(Figure in Northern Spain)
Suitable for high density areas feeding a central collection point.

6. Disposal: Today the disposal of wastes by landfilling is the ultimate fate of all solid waste.



SOLID WASTE OPEN DUMPSITE



SANITARY LANDFILL

THE END